

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(rvest)
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024'
durham_lwsp <- read_html(url)
durham_lwsp

## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
the_water_system_name <- durham_lwsp %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
the_water_system_name
```

```
## [1] "Durham"
```

```
the_contact_person <- durham_lwsp %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
the_contact_person
```

```
## [1] "Reginald Hicks"
```

```
the_ownership <- durham_lwsp %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
the_ownership
```

```
## [1] "Municipality"
```

```
the_max_day_use <- durham_lwsp %>%
  html_nodes("th~ td+ td , th~ td+ td") %>%
  html_text()
the_max_day_use
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4
month <- c("Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov", "Apr", "Aug", "Dec")

df_withdrawal <- data_frame(
  WaterSystemName = rep(the_water_system_name, 12),
  ContactPerson = rep(the_contact_person, 12),
  Ownership = rep(the_ownership, 12),
  Month = month,
  MaxDayUse_MGD = as.numeric(the_max_day_use) %>%
  mutate(Year = 2024, Date = as.Date(paste(Year, Month, "01"), format = "%Y %b %d")) %>%
  mutate(Month = factor(Month,
    levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
      "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))) %>%
  arrange(Date)
```

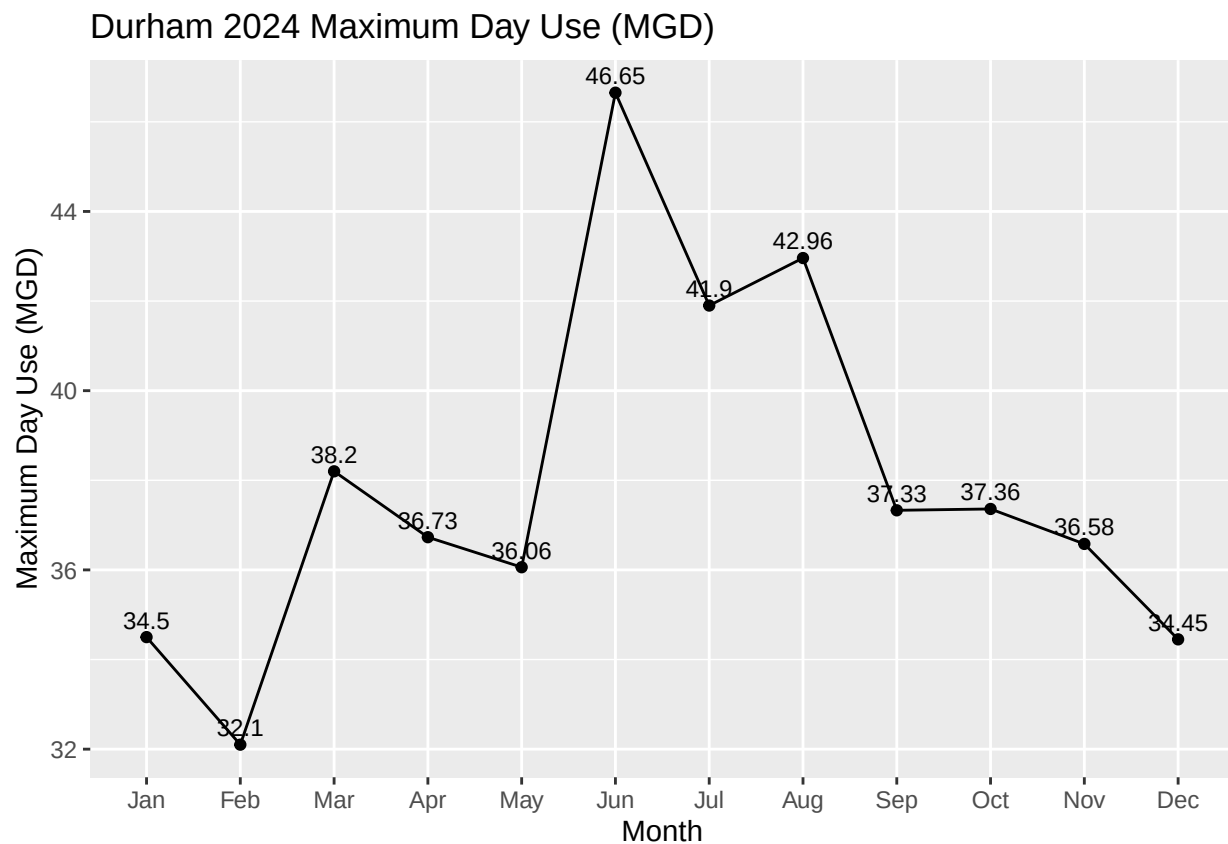
```
## Warning: 'data_frame()' was deprecated in tibble 1.1.0.
## i Please use 'tibble()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
df_withdrawal
```

```
## # A tibble: 12 x 7
##   WaterSystemName ContactPerson Ownership Month MaxDayUse_MGD Year Date
##   <chr>           <chr>         <chr>   <fct>    <dbl> <dbl> <date>
## 1 Durham          Reginald Hicks Municipa~ Jan      34.5  2024 2024-01-01
## 2 Durham          Reginald Hicks Municipa~ Feb      32.1  2024 2024-02-01
## 3 Durham          Reginald Hicks Municipa~ Mar      38.2  2024 2024-03-01
```

##	4	Durham	Reginald Hicks Municipa~	Apr	36.7	2024	2024-04-01
##	5	Durham	Reginald Hicks Municipa~	May	36.1	2024	2024-05-01
##	6	Durham	Reginald Hicks Municipa~	Jun	46.6	2024	2024-06-01
##	7	Durham	Reginald Hicks Municipa~	Jul	41.9	2024	2024-07-01
##	8	Durham	Reginald Hicks Municipa~	Aug	43.0	2024	2024-08-01
##	9	Durham	Reginald Hicks Municipa~	Sep	37.3	2024	2024-09-01
##	10	Durham	Reginald Hicks Municipa~	Oct	37.4	2024	2024-10-01
##	11	Durham	Reginald Hicks Municipa~	Nov	36.6	2024	2024-11-01
##	12	Durham	Reginald Hicks Municipa~	Dec	34.4	2024	2024-12-01

```
#5
ggplot(df_withdrawal, aes(x = Month, y = MaxDayUse_MGD, group = 1)) +
  geom_line() +
  geom_point() +
  geom_text(aes(label = round(MaxDayUse_MGD, 2)), vjust = -0.5, size = 3) +
  labs(
    title = "Durham 2024 Maximum Day Use (MGD)",
    x = "Month",
    y = "Maximum Day Use (MGD)"
  )
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)

- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.
scrape_lwsp <- function(PWSID, year){

  url <- paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
                PWSID, "&year=", year)

  the_website <- read_html(url)

  the_water_system_name <- the_website %>%
    html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
    html_text()
  the_contact_person <- the_website %>%
    html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
    html_text()
  the_ownership <- the_website %>%
    html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
    html_text()
  the_max_day_use <- the_website %>%
    html_nodes("th~ td+ td , th~ td+ td") %>%
    html_text()

  # month order as scraped from that table (matches your MGD scraping)
  month <- c("Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov", "Apr", "Aug", "Dec")

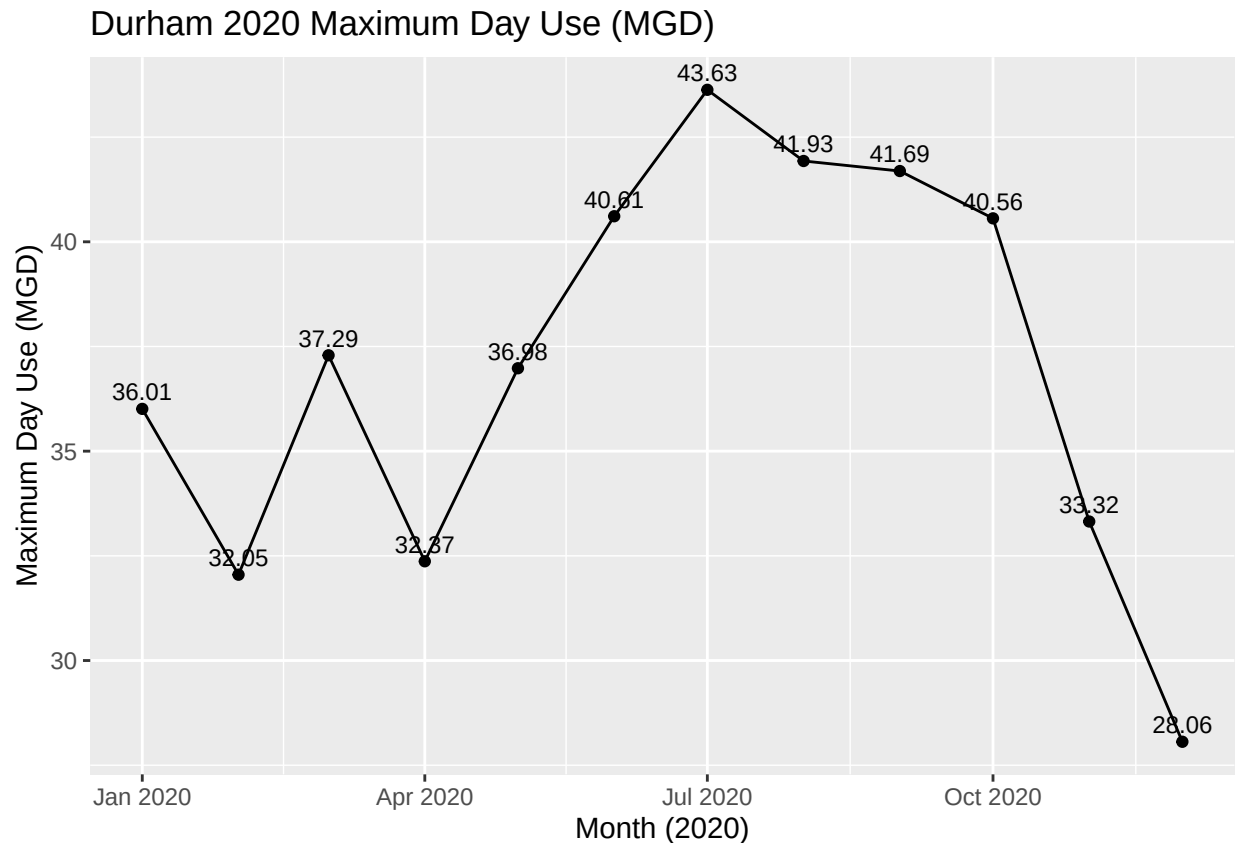
  df_withdrawal <- tibble(
    Month = month,
    Year = rep(year, 12),
    PWSID = rep(PWSID, 12),
    MaxDayUse_MGD = as.numeric(the_max_day_use),
    Water_System_Name = rep(the_water_system_name, 12),
    Contact_Person = rep(the_contact_person, 12),
    Ownership = rep(the_ownership, 12)
  ) %>%
  mutate(
    Date = as.Date(paste(Year, Month, "01"), format = "%Y %b %d"),
    Month = factor(Month, levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                                     "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))
  ) %>%
  arrange(Date)

  return(df_withdrawal)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
durham_2020 <- scrape_lwsp("03-32-010", 2020)
```

```
ggplot(durham_2020, aes(x = Date, y = MaxDayUse_MGD)) +
  geom_line() +
  geom_point() +
  geom_text(aes(label = round(MaxDayUse_MGD, 2)), vjust = -0.5, size = 3) +
  labs(
    title = "Durham 2020 Maximum Day Use (MGD)",
    x = "Month (2020)",
    y = "Maximum Day Use (MGD)"
  )
```



```
glimpse(durham_2020)
```

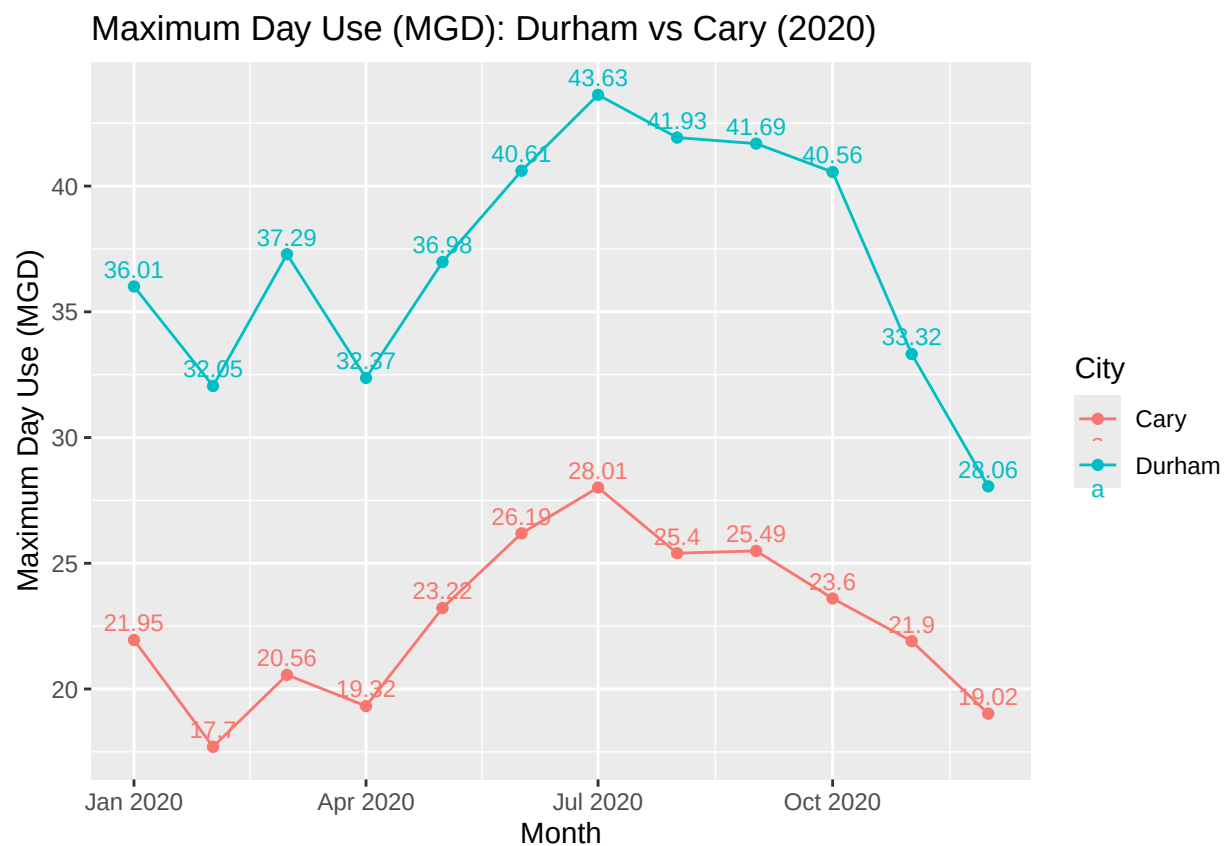
```
## Rows: 12
## Columns: 8
## $ Month      <fct> Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, No~
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ PWSID      <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ MaxDayUse_MGD <dbl> 36.01, 32.05, 37.29, 32.37, 36.98, 40.61, 43.63, 41.~
## $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ Contact_Person <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hicks"~
## $ Ownership   <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date        <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 202~
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data

with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
cary_2020 <- scrape_lwsp("03-92-020", 2020)
combined_2020 <- bind_rows(durham_2020, cary_2020) %>%
  mutate(
    City = case_when(
      PWSID == "03-32-010" ~ "Durham",
      PWSID == "03-92-020" ~ "Cary")
  )

ggplot(combined_2020, aes(x = Date, y = MaxDayUse_MGD, color = City, group = City)) +
  geom_line() +
  geom_point() +
  geom_text(aes(label = round(MaxDayUse_MGD, 2)), vjust = -0.5, size = 3) +
  labs(
    title = "Maximum Day Use (MGD): Durham vs Cary (2020)",
    x = "Month",
    y = "Maximum Day Use (MGD)",
    color = "City"
  )
)
```



```
glimpse(combined_2020)
```

```
## Rows: 24
```

```
## Columns: 9
## $ Month      <fct> Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, No~
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ PWSID      <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ MaxDayUse_MGD <dbl> 36.01, 32.05, 37.29, 32.37, 36.98, 40.61, 43.63, 41.~
## $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ Contact_Person <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hicks"~
## $ Ownership   <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date        <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 202~
## $ City        <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
```

9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

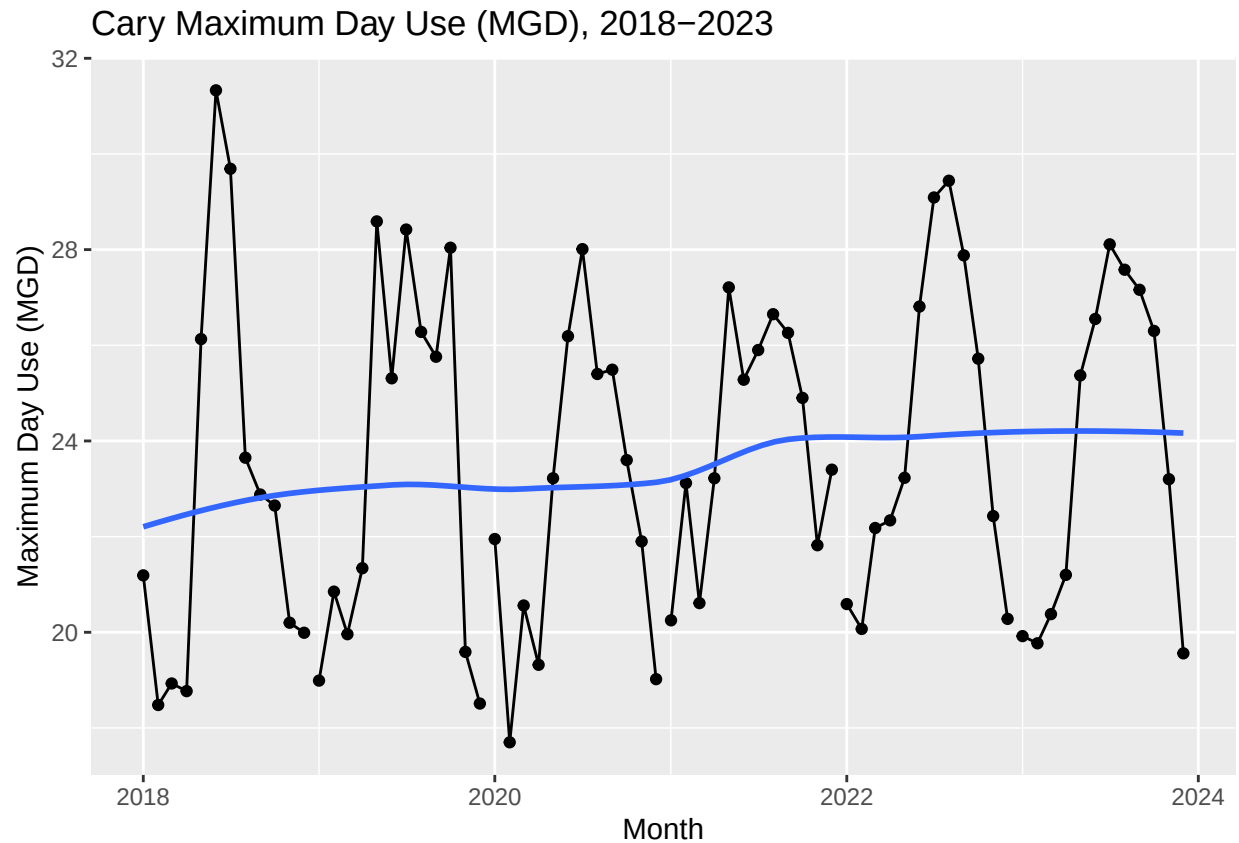
TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9

the_years <- 2018:2023
the_pwsids <- rep.int("03-92-020", length(the_years))
dfs_cary <- map2(the_pwsids, the_years, scrape_lwsp)
cary_all <- bind_rows(dfs_cary)

ggplot(cary_all, aes(x = Date, y = MaxDayUse_MGD, group = Year)) +
  geom_line() +
  geom_point() +
  geom_smooth(aes(group = 1), method = "loess", se = FALSE) +
  labs(
    title = "Cary Maximum Day Use (MGD), 2018-2023",
    x = "Month",
    y = "Maximum Day Use (MGD)"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
glimpse(cary_all)
```

```
## Rows: 72
## Columns: 8
## $ Month      <fct> Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, No~
## $ Year       <int> 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018~
## $ PWSID      <chr> "03-92-020", "03-92-020", "03-92-020", "03-92-020", ~
## $ MaxDayUse_MGD <dbl> 21.19, 18.48, 18.93, 18.77, 26.13, 31.33, 29.69, 23.~
## $ Water_System_Name <chr> "Cary", "Cary", "Cary", "Cary", "Cary", "Cary", "Car~
## $ Contact_Person <chr> "Jamie Revels", "Jamie Revels", "Jamie Revels", "Jam~
## $ Ownership   <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date        <date> 2018-01-01, 2018-02-01, 2018-03-01, 2018-04-01, 201~
```

Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes. By looking the smoothed line of the plot, from 2018 to 2023, the overall water usage is increasing from around 22 MGD to over 24 MGD. Even it shows an obvious seasonality on the water usage trend, the water usage increases indeed over time in Cary. >